VIT-AP UNIVERSITY

Summer Internship Project

Zeroth Review: Problem Identification

Team:

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Title of the Project

InstantB2C – Web Platform for Micro and Home-based B2C Sellers

Software Requirements

- **Operating System (Server)**: Linux or Windows 10+
- **Database**: MySQL with Sequelize ORM
- **Frontend**: Next.js (React) with Tailwind CSS & MUI
- **Backend**: Node.js, Express.js (via Next.js API Routes), Python
- **Authentication**: JOSE JWT-based Auth
- **AI Integration**: Model Context Protocol (MCP) for intelligent bundling and seller prompts

Hardware Requirements

- **Processor**: Intel Core i5 or higher
- **RAM**: 8 GB
- **Storage**: 50 GB minimum

Abstract

InstantB2C is a Software as a Service (SaaS) web platform tailored to empower micro and small business owners—especially home-based entrepreneurs and social commerce sellers—to seamlessly set up and manage their own personalized online storefronts. Whether it's home-based bakers, sweet makers, or boutique sellers operating on platforms like Instagram or WhatsApp, InstantB2C offers a centralized, intuitive, and tech-free experience to list products, share store links, manage customer orders, and automate fulfillment workflows.

This project reimagines traditional e-commerce by eliminating technical barriers, offering plug-and-play tools for sellers to establish and scale their digital presence in minutes.

To further enhance business decision-making, InstantB2C integrates AI capabilities using the Model Context Protocol (MCP) — a standardized format for

structured interactions with AI models — to power intelligent product bundling and sales recommendations.

Project Idea and Novelty

InstantB2C addresses a major gap in the current e-commerce ecosystem: the lack of simple, intelligent digital tools for small and home-based sellers who operate primarily through social platforms like WhatsApp or Instagram. These sellers often face three main challenges:

1. Lack of technical knowledge to build a store
2. No access to real-time analytics or AI-based suggestions
3. Manual, unscalable workflows (e.g., Excel sheets, DMs for orders)

What Makes InstantB2C Novel?

- Plug-and-play store creation: Sellers can launch a personalized online storefront in minutes—no coding or tech skills required.
- Unified Operations Hub: Product listing, order tracking, inventory, promotions, and loyalty programs all in one dashboard.
- On-the-go Business Intelligence:
    - AI-powered product bundle suggestions (e.g. sweets combos that typically sell together).
    - Real-time business prompts — "Add more of this flavor–it's trending this week."
- No-Customer-Registration Checkout: Unlike large platforms, InstantB2C doesn't require customers to create accounts, making it ideal for casual or one-time buyers common in home-based commerce.
- Built-In Order Management & Discounts: Sellers can easily manage orders, track statuses, and apply promotions — all from a unified dashboard.

What It Brings to the Table

Feature	Benefit
Modular design	Easy to extend with payments, loyalty, etc.
AI-assisted workflows	Helps sellers make better decisions with less guesswork

Real-time micro analytics	Tailored insights for solo/small team sellers
End-to-end stack	One system for storefront, backend, and analytics
No technical entry barrier	Designed specifically for non-technical sellers

In short, InstantB2C bridges the digital divide for offline and micro-sellers by combining modern web technologies with accessible AI tools, making it a powerful yet approachable business assistant.

Modules

1. Seller/Admin Panel

- Product Management
- Inventory Handling
- Smart Product Bundling (MCP-powered AI)
    - Uses MCP to structure recent order data and send it to an AI model (e.g., GPT-4)
    - AI suggests frequently bought-together items in a standardized response
    - Displayed in the admin panel with suggested combos
- Order Dashboard
    - Orders view
    - Order Status Management (Pending → Packed → Ready to Ship → Delivered)
- Data Analytics (MCP-powered AI prompts)
    - Generates weekly sales insights and alerts using structured MCP format
    - AI provides prompts like:
        "Add more Mango Barfi — it’s trending this week!"

2. Customer Interface

- Product Browsing by Category
- Add to Cart and View Cart
- Simple Checkout Form (name, contact, delivery address)

- Order Summary + Price Calculation (with dynamic weight and discount handling)
- No customer registration required

3. Discount & Promotions Module

- Create and manage time-bound product discounts
- Automatically applies the best applicable discount during checkout

4. Extensions

- Loyalty System & Referrals (coins/coupons)
- Delivery Status Page (with manual updates)
- Payment Gateway Integration (UPI, Razorpay)

5. Authentication and Security

- Secure login for admins using JOSE JWT
- Role-based access control (Admin, Delivery, Read-only)
- Input validation on both client and server
- Optional 2FA integration (future enhancement)

Conclusion

InstantB2C is a complete digital storefront and operations platform designed for micro and home-based B2C sellers. With a no-code setup experience and a smooth customer-facing UI, it empowers traditional offline sellers to digitize their business with minimal effort.

By integrating AI-driven insights using the Model Context Protocol (MCP), InstantB2C not only simplifies store setup and management, but also intelligently supports business growth through automated product bundling suggestions and data-driven selling prompts — making e-commerce more accessible and strategic for non-technical sellers.